# Bubble Size and Bubble Velocity Distribution in Bubble Columns under Industrial Conditions

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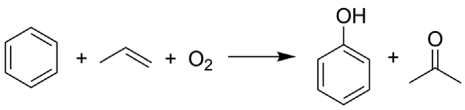
Bubble column reactors are widely used in many industrial applications due to their simplicity and safety of operation. Despite these advantages, the design and scale-up of bubble column reactors is still challenging especially for industrial conditions at elevated pressure and temperature. One reason is the uncertainties concerning the specific interfacial area which is directly dependent on the bubble size distribution, bubble velocity, and gas hold-up. All these parameters are difficult to measure under industrial conditions due to the opaqueness of the bubbly flow and the safety risks of using organic solvents at elevated pressures and temperatures. This article introduces endoscopic bubble image velocimetry, a new measuring method that enables the detection of bubble sizes and bubble velocities in organic solvents at elevated pressure and temperature ($p_{max} = 1.85$ MPa and $T_{max} = 70$ °C) for maximal gas hold-ups of 16 %. For this system it becomes evident that the bubble size distribution for low superficial gas velocities is almost unaffected by pressure and temperature, whereas the bubble velocity decreases slightly.

**Keywords:** endoscopic bubble image velocimetry, photo-optical measurement technique, elevated pressure

## INTRODUCTION

Bubble column reactors, an example of multiphase reactors, have been proven effective for a variety of chemical reactions and thus, are widely used in the chemical and biochemical industries. Typical reactions are oxidation, hydrogenation, chlorination, phosgenation, hydroformylation, and alkylation as well as fermentation processes.[1] For instance the Hock process, an example of an oxidation reaction according to the following:




(1)

is usually applied in bubble column reactors with a large annual production of 1870 kt phenol (INEOS Phenol). Despite frequent application, the complex flow conditions and the enormous number of coupled parameters, such as gas and liquid velocity, as well as bubble diameter and velocity, which determine the mass transfer between the liquid and gaseous phase, are still hardly predictable.[2] Although there are measurement techniques available based on conductivity, electro- and photo optics, as well as ultrasound[3–4] it is difficult to gain information about bubble diameter and velocity of bubbly flows with relevance for industrial use. One of the reasons for this is the lack of measurement techniques applicable under industrial conditions, in other words organic solvents under higher pressure and temperature.

Investigations which have already been conducted are partially controversial with regard to the quantitative effect of pressure (see Table 1). While it is widely agreed that the bubble size becomes smaller due to the compressibility of the gaseous phase and that both coalescence and break-up are affected, it remains undecided above which pressure the bubble size is indeed influenced. A decrease in bubble velocity is to be expected, but it is rarely investigated. Therefore more detailed information on bubbles concerning their size, shape, and movement under elevated pressure/temperature in organic solvents is required.

This paper provides new insights into a pilot reactor at the Evonik Industries AG, Marl by means of the endoscopic measurement technique. As continuous phase the organic solvent cumene is used while nitrogen is supplied as the dispersed phase. The pressure/temperature range has been adapted to 0.1–3.6 MPa and 25–70 °C. The measurement technique has been applied to a maximum gas hold-up of 16 %, which corresponds to a superficial gas velocity of 0.052 $m \cdot s^{-1}$. Due to the robust design of the EPIV probe, very high superficial liquid velocities can be applied.

## THE STATE OF THE ART

The bubble size distribution (BSD) in bubble columns is affected by bubble formation, bubble coalescence and break-up, mass transfer, and expansion due to the decreasing pressure with increasing rising height.

Bubble formation is divided into bubble gassing and jet gassing depending on superficial gas velocity and is illustrated in Figure 1.[5]

At low superficial gas velocities ($We < 2$), single bubbles are formed at the orifice. Single bubbles or bubble chains characterize the often mentioned bubble gassing. Moreover, almost no

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**Table 1**. Overview about research concerning BSD under industrial conditions

| Authors | Physical system (gas/liquid) | Experimental conditions (pressure/temperature) | Geometry |
|---|---|---|---|
| Pohorecki et al.[11] | Nitrogen/cyclohexane | 0.1–1.1 MPa, 30–160 °C | D = 0.30 m, H = 4.00 m |
| Lemoine et al.[34] | Nitrogen, air/toluol, toluol-mixtures | 0.2–0.8 MPa, 26.85 °C | D = 0.316 m, H = 2.8 m |
| Schäfer et al.[23] | Nitrogen/water, cyclohexane, cyclohexanone, cyclohexanol, and ethanol | 0.1–4.6 MPa, up to 175 °C | D = 0.2 m, H = 1 m |
| Kang et al.[20] | Air/aqueous CMC solutions | 0.1–0.6 MPa, ambient | |
| Lin et al.[28] | Nitrogen/Paratherm NF | 0.1–15.2 MPa, 27–78 °C | D = 0.05 m, H = 0.8 m & D = 0.1 m, H = 1.58 m |
| Letzel et al.[21] | Nitrogen/water | 0.1–1.3 MPa, ambient | D = 0.15 m |
| Kojima et al.[22] | Oxygen, nitrogen/tap water, citric acid solution, aqueous solution of glucose oxidase | 0.1–1.1 MPa | D = 0.045 m, H = 0.9–1.2 m |
| Wilkinson et al.[26] | Nitrogen/water, *n*-heptane, ethylen glycole, sodium sulphite solution | 0.1–1.5 MPa, 20 °C | D = 0.15 m, H = 1.5 m |
| Soong et al.[25] | Nitrogen/drakeol-10 | 0.1–1.36 MPa, 20–265 °C | D = 0.1 m |
| Idogawa et al.[19] | Hydrogen, nitrogen/water, methanole, acetone, aquous alcohol solution | 0.1–5 MPa | D = 0.05 m |

coalescence and break-up occur. Therefore the primary BSD right behind the gas distributor represents for the most part the distribution for the entire bubble column reactor.[5]

Based on force equilibrium for the static bubble formation on a sieve plate between buoyancy, gravity, and interfacial tension force the primary bubble size can be assessed as follows:

$$d_b = \sqrt[3]{\frac{6\, d_o\, \sigma}{\Delta\rho\, g}} \tag{2}$$

according to Mersmann.[6]

With increasing superficial gas velocity, bubbles are bouncing and merging to a jet. With some distance from the gas distributor, single bubbles are finally formed due to the turbulent flow forces (compare Figure 1).[7] Thus bubble size and consequently velocity can differ strongly and a distribution needs to be considered. However, the described mechanisms are valid for Newtonian and only partly for non-Newtonian liquids.

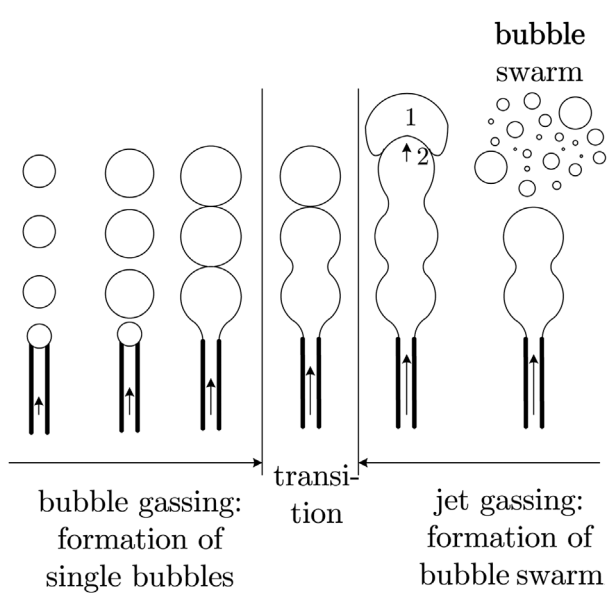


**Figure 1**. Bubble gassing and jet gassing in bubble columns.[5]

Brauer[5] distinguishes between bubble and jet gassing with the help of the critical parameter $\Phi_{crit}$:

$$\Phi = 1 + \sqrt{1 + \frac{We^2}{Fr}} \tag{3}$$

with respect to Weber number:

$$\mathrm{We} = \frac{w_o^2 d_o \rho_g}{\sigma}$$

as well as Froude number:

$$Fr = \frac{w_o^2}{d_o g}$$

As can be seen in Figure 2 the critical value of $\Phi_{crit} = 27$ is determined for the regime change from bubble to jet gassing by the application of the dimensionless bubble diameter:

$$d^* = \frac{d_b}{\sqrt[3]{\frac{3 \cdot \sigma \cdot d_o}{\rho_l \cdot g}}} \tag{4}$$

versus $\Phi$.

If coalescence and break-up come into play, the calculation of a characteristic bubble diameter from the BSD is necessary. Therefore the Sauter mean diameter (SMD), as follows:

$$d_{S32} = \frac{\sum n_i d_i^3}{\sum n_i d_i^2} \tag{5}$$

is most often applied.[8]

Different correlations for determining the SMD have been developed. A frequently used one is as follows:

$$\frac{\mathbf{d_{BS}}}{\mathbf{d}} = \mathbf{26}\left(\frac{d_{S32}}{D} = 26\left(\frac{gD^2\rho_l}{\sigma}\right)^{-0.5} \cdot \left(\frac{gD^3}{\nu_l^2}\right)^{-0.12} \cdot \left(\frac{w_0^g}{\sqrt{gD}}\right)^{-0.12}, \tag{6}$$



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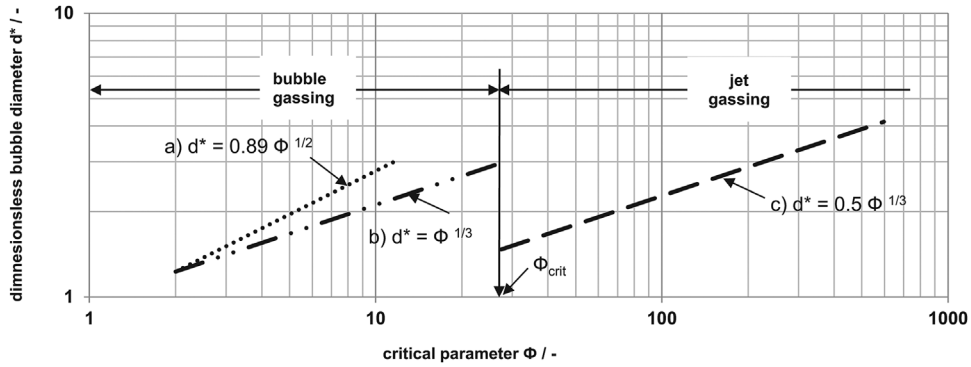


**Figure 2.** Plot of the dimensionless bubble diameter versus the critical parameter in order to differentiate into bubble gassing and jet gassing. Correlations (a–c) are based on experimental data: correlation (a) or (b) for bubble gassing depending on material system and frequency of the bubble formation, correlation (c) for jet gassing.[5]

which was developed by Akita and Yoshida[9] using different material systems.

In his 1991 thesis Wilkinson [10] published the following equation:

$$d_{S32} = \sqrt{\frac{8.8 \cdot \sigma}{g \cdot \rho_l} \cdot \left(\frac{w_0^g \cdot \eta_l}{\sigma}\right)^{-0.04} \cdot \left(\frac{\sigma^3 \cdot \rho_l}{g \cdot \eta_l^4}\right)^{-0.12} \cdot \left(\frac{\rho_l}{\rho_g}\right)^{0.22}} \quad (7)$$

based on experiments with not only water but also mono-ethylene glycol, cyclohexane, and $n$-heptane. Nitrogen is used as the gas phase in most of the experiments. Other gases are helium, argon, and carbon dioxide. Experiments were conducted in a reactor with 0.15 m inner diameter at ambient temperature and two pressures of 0.1 MPa and 1.5 MPa.

While the two correlations in Equations (6) and (7) were fitted to a dimensional analysis using experimental data, Pohorecki et al.[11] used a different approach. Their equation is based on the simplified population balance by Price and Blanch that assumes bubble break up leads always to two bubbles with the same size, and coalescence means the integration of two smaller bubbles into a larger one.[12] Furthermore, both procedures are expected to frequently occur at the same time. Pohorecki et al.[11] validated their model, as follows:

$$d_{S32} = 0.289 \cdot \rho_l^{-0.552} \cdot \eta_l^{-0.048} \cdot \sigma^{0.442} \cdot w_0^{g-0.124} \quad (8)$$

using different reactor geometries (including one with an inner diameter of 0.304 m and a length of 3.99 m), operation conditions (temperature up to 160 °C, pressure up to 1.1 MPa), and material systems (for example cyclohexane/nitrogen). Additional correlations are summarized by Pohorecki et al.[11] Further methods to model BSDs have been developed by using population balance equations (PBE). However, this topic is beyond the focus of this paper; the reader is referred to the literature.[13–17]

The influence of mass transfer on BSD is not addressed in this paper, because all phases in the experiments are assumed to be in equilibrium. Furthermore the influence of the expanding bubble by increasing rising height and decreasing pressure due to the hydrostatic height becomes comparably small under high pressure conditions and should be negligible.

Since elevated pressure and temperature are of great interest for industrial applications, different investigations have been conducted to study their influence on the most important parameters, but there are still contrary statements.[18] Focusing on the gas hold-up, as one of the most significant parameters, it is commonly agreed that the gas hold-up is raised with an increase in pressure.[18] Indeed, a change in gas hold-up is attributable to the hydrodynamic parameters, bubble size, and velocity. Table 1 shows the investigations in bubble columns under industrial conditions, or, in other words, elevated pressure and temperature, when the bubble sizes are also considered. In comparison to the bubble size, the bubble velocity under elevated pressure is rarely investigated.[18]

Several studies have shown that the bubble sizes are smaller at elevated pressures.[19–23] Due to smaller initial bubbles formed at the gas distributor some of the authors conclude that there is a decrease in bubble size and a narrower BSD.[19,20] Moreover, a different behaviour in coalescence and fragmentation processes for higher pressure is observed. Letzel et al.[21] pointed out that a larger gas density leads to higher instabilities and, in consequence, more bubble breakup and less bubble coalescence occurs. This is also confirmed by Kojima et al.[22] when using different material systems. Not only for higher pressure, but also for an increased temperature, Schäfer et al.[23] observed less bubble coalescence, too. It is therefore to be expected that the BSD shifts to smaller bubbles.

However, some authors come to the conclusion that the BSD does not change with increasing pressure.[19,24–27] For low gas velocities ($We_g < 0.1$), for example Wilkinson[10] determined that pressure has no influence on bubble formation due to a low gas momentum.

Apart from the BSD, it is essential to know the bubble velocity, because it determines residence time and particularly the gas hold-up. There is a general agreement that bubble rise velocity is reduced in the case of elevated pressure (e.g. Letzel et al.,[21] Lin et al.[28]). The bubble velocity derived from the force equilibrium, as follows:

$$w_b = \sqrt{\frac{4}{3}\frac{d_b g \Delta\rho}{c_d \rho_l}} \quad (9)$$

is proportional to the square root of the density difference.[5] Whereas the liquid density remains almost constant according to the incompressibility, the gas density increases with increasing pressure. Consequently, the density difference between liquid and gas and therefore the buoyancy force becomes smaller and leads to a reduced bubble velocity.



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In order to determine bubble size and velocity, there are already several measurement techniques available based on conductivity, electro-optics, and photo-optics as well as ultrasound.[3,4] Well established are, for example, needle probes that use the time difference between signals (impedance, conductivity, light reflection) caused by the bubble interface if the bubble is penetrated by the needle. Despite the fact that this is a very fast and universal opportunity to get information about bubble sizes and velocities, the position of the penetration through the bubble is unknown and therefore the accuracy is limited. More precise are needle probes with multiple fibres but some uncertainty concerning the bubble shape remains.[29] With the purpose of getting information of the bubble shape, photo-optical measurement methods can be applied by using a camera which takes images of the bubbly flow. For images taken from outside the bubble column, only the flow near the wall is recorded. Besides the information of the bubble size, bubble velocity can be obtained by taking pictures in a double shutter mode, using the offset of the balance point and the time difference. Compared to the conductivity and electro-optical probes, this method is non-invasive. However, it is limited to lower gas hold-ups. Thus for measuring at higher gas hold-ups a camera connected to the endoscope, which goes directly into the bubbly flow, can be applied for measuring within the bubble column. Maaß[30] has already shown the application of this principle method in stirred tanks and Mickler[31] in the field of extraction columns while within this paper the measurement method will be applied in bubble columns. More recently measurements with ultra-fast X-ray tomography for three-dimensional measurements of bubble sizes and positions within the column volume have been conducted.[32] Further details about measurement techniques can be found for example in Tropea.[33]

The literature survey shows that many investigations have been done to determine the influence of elevated pressures on gas hold-up in bubble columns. Nevertheless the published results are often contradictory and difficult to understand because of a lack in local information about the bubble properties such as BSD and bubble velocity distribution. In this paper a measurement technique will be presented which allows the determination of these parameters. This enables an insight into the behaviour of these essential parameters under industrial conditions, i.e. for an organic solvent under elevated pressure and temperature.

## EXPERIMENTAL SETUP

### Equipment

Experiments are conducted at Evonik Industries AG (Marl, Germany) in a bubble column made of stainless steel with a height of 4 m (liquid level: 3.88 m) and a diameter of 0.33 m. Operating conditions are pressures of up to 3.6 MPa[1] and temperatures up to 70 °C. The temperature has been changed in a production relevant range to investigate the influence of the temperature on viscosity and hydrodynamics. Because the effect of temperature changes is low in comparison to other influences, this parameter will not be in the focus of this article.

[1]Experiments were conducted up to 3.6 MPa for very low superficial gas velocities. However, results are just shown up to 1.85 MPa to enable the comparison of different operating pressures for the same superficial gas velocities.

In Figure 3, a simplified flowchart of the industrial bubble column is shown. Nitrogen, as the gas phase, is provided by a compressor. A backpressure regulator at the gas outlet is used to adjust the pressure between 0.1 and 3.6 MPa. With the aid of a heater, the inlet gas temperature is between 21–23 °C. After being measured by a gas flowmeter (Krohne, H250), the gas flow is distributed by a perforated plate sparger (352 holes with a diameter of 1 mm). In the investigation shown here, cumene is used as the liquid phase and circulated by a pump. The liquid phase is operated in an upward co-current flow and flow rate is also measured by a flowmeter (Krohne H250). Before entering the column, the liquid phase is heated up to the particular operating temperature by using a heat exchanger. At the top of the column the gas and liquid phase are separated in a buffer vessel. As can be seen in Figure 3 pressure sensors are installed along the reactor. From the pressure difference the gas hold-up can be calculated according to the following:[35]

$$\varepsilon_g = 1 - \frac{\Delta p}{\Delta p_0} \tag{10}$$

$\Delta p_0$ denotes the static pressure of the liquid column without gas, whereas $\Delta p$ is measured for the dispersed state of the two-phase column.

Various measurement techniques are applied in the reactor[2,36] shown in Figure 3, this study, however, focuses on the photo-optical endoscopic technique. Endoscopes required for this investigation are positioned at 2.8 m above the gas distributor and will be explained in detail in the next section.

### Measurement Technique

For the investigation of the bubbly flow a photo-optical endoscopic setup is used. In this way it is possible to gain useful information about the two-phase flow directly within the reactor for higher gas hold-ups ($\varepsilon_g$ up to 16 %) compared to a photo-optical acquisition from outside. The principle is based on the classical particle image velocimetry (PIV), whereby particles are distributed in the liquid phase and follow the flow without inertia. By taking two images within a defined time distance and due to the offset of particles from one picture to the next one, velocity can be obtained. When the same procedure is applied to bubbles, it is called endoscopic bubble image velocimetry (EBIV).

For this photo-optical acquisition an endoscope by Storz is combined with a 100 mm Canon lens. The images are acquired with a digital sCMOS camera (ILA.PIV.sCMOS) with double shutter mode. One image pair (base image (A) and cross image (B)) is taken within a short time frame of, for example, 1 ms. The camera is configured to take ten of those image pairs per second. The bubble size is determined by the base image (A). In order to obtain the bubble velocity, the offset of the centre of area is calculated from the base image (A) to the cross image (B) (compare Figure 4).

For the acquisition of the bubbles a field of view with a diameter of 8 mm is used. The resolution of the images is 2560 × 2160 pixels. At every point of measurement 5000 image pairs are taken. Since the number of image pairs is fixed, the number of detected bubbles differs between points of measurement, although there are always more than 500 detected bubbles, which is the minimum number of bubbles as described by Deckwer.[37] To ensure that the BSD is representative, the cumulative mean value is considered so that the number of analyzed bubbles can be regarded as sufficient.



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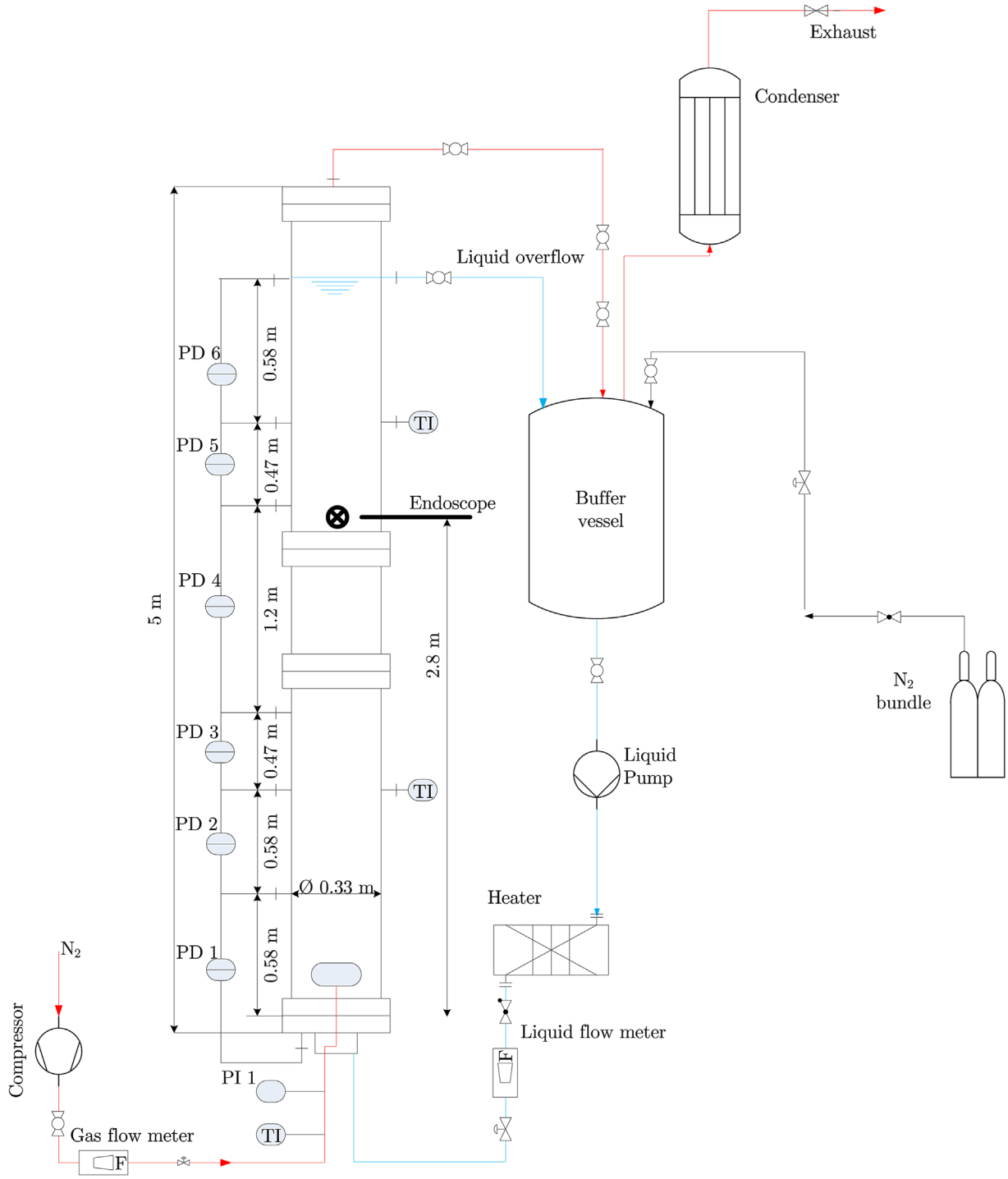


**Figure 3**. Sketch of the industrial bubble column at Evonik Industries AG (Marl, Germany).

For the illumination a pulsed laser system is applied. The Intelligent Laser Application GmbH (ILA, Juelich) developed a special laser endoscope for industrial conditions for pressures up to 10 MPa and a maximum temperature of 150 °C. The laser endoscope is implemented at a 90° angle to the camera endoscope, which is in turn implemented in a jacket tube, as can be seen in Figure 4. The light of a dual Nd:YAG laser (Quantel Evergreen 200), which generates a wavelength of 532 nm, is connected to the laser endoscope by using a light guiding arm.

Since the direct reflection of the laser light on the bubbles can destroy the CCD chip, a direct illumination is not possible. Hence, for an indirect illumination, the fluorescent dye Nile red (3 mg · $L^{-1}$) is dissolved in the liquid phase. In order to find the appropriate dye, different dyes are tested. However, for cumene, as an organic solvent, just Nile red is appropriate for the used spectrum of light. It was ensured before the experiments that surface tension is not affected by the solved dye by using a tensiometer (Krüss, BP50). Concentration has been chosen with respect to the illumination quality to enable a feasible automatic



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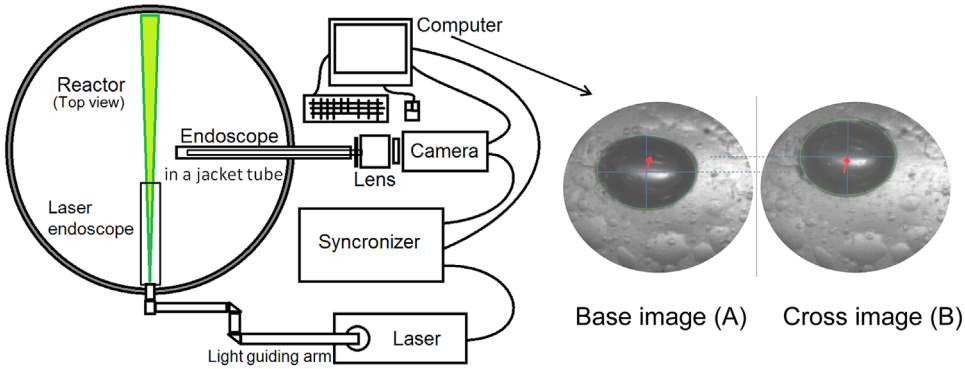


**Figure 4.** Schematic setup of the endoscopes in the reactor (left) image pair of an analyzed bubble (right).

analysis. The fluorescence of the dye is activated by using the laser light. In this way a uniform background illumination is realized. To protect the camera sensor from intensive laser light, a filter is implemented which selects only the emitted light by the fluorescence dye. When excited with laser light with a wavelength of 532 nm, Nile red fluoresces around 590 nm, therefore a bandpass filter with a centre wavelength of 590 nm and a half-power bandwidth of 20 nm is used to block the direct laser light and only let the fluorescence pass through to the sCMOS sensor. The laser illuminates the centre of the bubble column and the focus level of the camera is placed in the same position. A synchronizer from ILA GmbH is used to trigger the camera with respect to the pulsed laser system.

## AUTOMATIC ANALYSIS

The software "Bubble Finder®" used for the automated analysis is an in-house development of the ILA GmbH, Jülich in cooperation with the Institute of Multiphase Flows, Hamburg University of Technology. The images of the bubbles are scanned by the automatic software in three steps: pre-processing, detection, and post-processing. Concerning the pre-processing a default proposal is made, but the researcher can change the setting according to the available images. The pre-processing involves noise removal schemes which are a combination of the following preprocessing routines:

- contrast enhancement;
- smoothing filters (Gaussian, simple average, median) with adjustable kernel size;
- morphology filters (dilate, erode, and also their common combinations: open, close, top hat, black hat) using one of the three kernels (rectangle, cross, ellipses), each with adjustable size.

The optimal settings and combination of the filters depend on the raw image quality and are defined for each dataset separately. Usually, the threshold has been set to 35 and a minimum roundness of around 0.6 as well as a maximum roundness of 1 have been used.

The detection step involves contour finding and ellipse fitting, whereas ellipses with diameter smaller than 7 pixels, however, are ignored, since they are very probably unfiltered noise. According to the calibration setup, this corresponds to 50 μm. The post-processing involves outlier removal. The outliers are determined from their size, roundness, and exclusiveness. For the velocity estimation the software detects the displacements of the bubbles by matching each bubble which appears in both the base (A) and cross (B) image. The bubbles are matched with respect to their similarity and their position, in other words, a best match should have the highest similarity value and a reasonable position difference/displacement. A reasonable position difference/displacement corresponds to the product of time difference and estimated velocity. The similarity computation is influenced by the roundness-difference and the area-difference. The base and cross image in Figure 4 show that the tracking algorithm for the detection of bubble velocities detects the vertical as well as the horizontal velocity separately. For this study only the vertical velocity is used, according to the definition of the rising velocity by Brauer.[5] Furthermore it must be pointed out that with the EBIV measurement technique the *absolute velocity* of the bubble is measured. This velocity includes the liquid velocity due to the net flow or buoyancy driven flow and mainly influences the gas hold-up. In contrast, for mass transfer the bubble velocity relative to the liquid phase has to be considered.[5] The relative velocity ($w_{bs,rel}$) can be obtained from the difference of the absolute bubble velocity ($w_{bs,abs}$) and the overlaid liquid velocity ($w_l$) according to the following:

$$w_{bs,rel} = w_{bs,abs} - w_l \tag{11}$$

by using averaged values in time and space. Because these measurements are not the focus of this paper, further details can be found in Bothe and Schlüter.[38]

The measurement errors due to this acquisition are reduced to a minimum, for example by using a large aperture (small f-number) in order to achieve a low depth of focus. Before conducting these measurements, this is confirmed theoretically and experimentally in the lab-scale. In addition the bubble velocities obtained from the endoscopic measurement are compared to the results from an electro-optical probe (B-POP probe, A2 Photonics), both in lab-scale. For superficial gas velocities up to $0.036\ \mathrm{m \cdot s^{-1}}$ the deviation of the absolute bubble velocity determined by EBIV and the electro-optical probe is 5.5 % on average. In contrast, the bubble size cannot be compared adequately due to the fact that just a chord length is determined by the optical probe.

The accuracy of this measurement method is also dependent on the automatic analysis. For quantifying this error, the difference between automatically and manually analyzed bubbles is considered. An error of 3.93 % is determined by analyzing the Sauter mean as well as the mean diameter of 217 bubbles. Since for one measurement point usually around 1000 bubbles are considered with the automatic system, the error will be more likely lower due to a larger number of bubbles.



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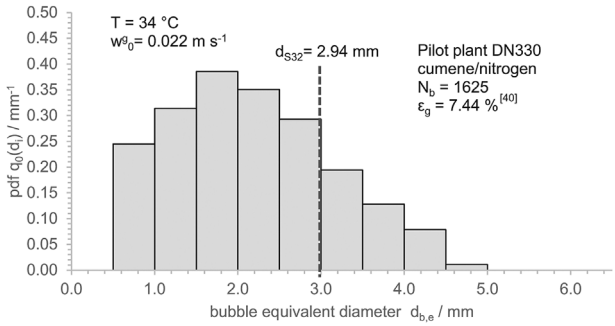


**Figure 5**. Bubble size distribution at p = 0.1 MPa, T = 34 °C with $w_0^g = 0.022\ \mathrm{m \cdot s^{-1}}$ and $w_0^l = 0.018\ \mathrm{m \cdot s^{-1}}$.

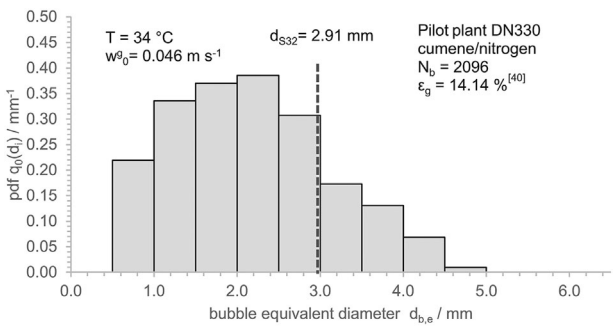


**Figure 6**. Bubble size distribution at p = 0.1 MPa, T = 34 °C with $w_0^g = 0.046\ \mathrm{m \cdot s^{-1}}$ and $w_0^l = 0.018\ \mathrm{m \cdot s^{-1}}$.

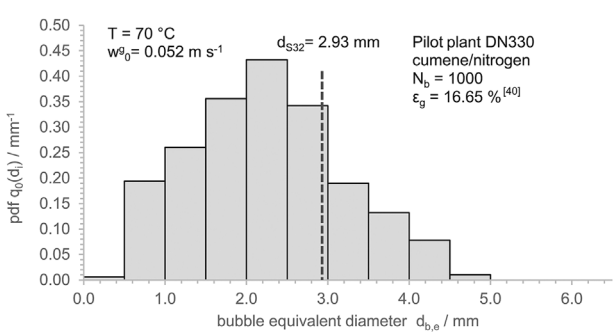


**Figure 7**. Bubble size distribution at p = 0.10 MPa, T = 70 °C with $w_0^g = 0.052\ \mathrm{m \cdot s^{-1}}$ and $w_0^l = 0.018\ \mathrm{m \cdot s^{-1}}$.

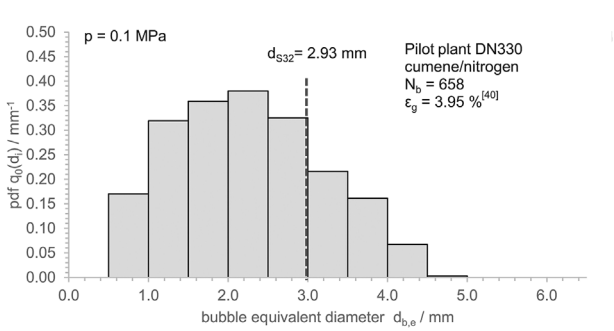


**Figure 8**. Bubble size distribution at p = 0.10 MPa, T = 69 °C with $w_0^g = 0.006\ \mathrm{m \cdot s^{-1}}$ and $w_0^l = 0.008\ \mathrm{m \cdot s^{-1}}$

## EXPERIMENTAL RESULTS AND DISCUSSION

In order to investigate the influence of the operating conditions, especially pressure, on the hydrodynamics of the gas-liquid flow, first the BSD is considered by using the probability density function (pdf). Furthermore, the SMD is experimentally determined according to Equation (5) and compared to the different correlations (Equations (6–8)). Afterwards the bubble velocity distribution for two different pressures is shown.

### Bubble Size Distribution

With the purpose of comparing the different bubble sizes, an equivalent diameter is required that is related to a sphere with the same volume or the same surface as the irregularly shaped bubble. Because most of the observed bubbles are close to an ellipsoidal shape, an ellipsoid of rotation can be assumed, which enables the applicability of the ellipse fitting algorithm in the Bubble Finder® software (compare *Automatic analysis*). In this way the deviation from the spherical shape is also taken into account. In this study the surface-equal diameter $d_{b,e}$ is applied to the detected bubbles according to the surface of the ellipsoid of rotation (S):

$$d_{b,e} = \sqrt{\frac{S}{\pi}} \tag{12}$$

because the interface is more significant for mass transfer performance than the bubble volume. After the detection, all bubbles of one measurement point are classified into bubble size classes with an interval of 0.5 mm. Experiments with the same operating conditions, conducted in two temporally different measurement campaigns, are compared and showed a deviation of around 8 %. For the sake of clarity, just one BSD will be shown in the diagrams.

Figures 5 and 6 show the BSD of the surface-equal diameter for a superficial liquid velocity of $0.018\ \mathrm{m \cdot s^{-1}}$, a temperature of 34 °C, and a constant pressure of 0.10 MPa. The superficial gas velocity is more than doubled from $0.022\ \mathrm{m \cdot s^{-1}}$ (Figure 5) to $0.046\ \mathrm{m \cdot s^{-1}}$ (Figure 6). It can be observed that no significant change in the SMD is detected, despite the fact that a light shift to smaller bubble sizes appears with increasing superficial gas velocity. This is not surprising because for the experiments the parameter $\Phi$ (see Equation (3)) is between $2.03 < \Phi < 3.02$ and therefore much lower than $\Phi_{crit} = 27$, suggesting bubble gassing. Moreover, since the BSD does not change for different operating conditions at the level of 2.8 m above the gas distributor, neither coalescence nor breakup processes seem to be relevant for these operating conditions.

BSD is determined for the same conditions apart from the temperature, which is raised to 70 °C (compare Figure 7). However, the increase in temperature has again no significant effect on the SMD. There is no difference in the bubble sizes above 3.0 mm, whereas a slight shift can be detected for smaller bubbles.

With the purpose of investigating the influence of pressure on BSD, the pressure was increased from 0.10 to 1.85 MPa at a temperature of about 70 °C for a superficial gas velocity of $0.006\ \mathrm{m \cdot s^{-1}}$ and liquid velocity of $0.008\ \mathrm{m \cdot s^{-1}}$ (see Figures 8 and 9).

If the two BSDs in Figures 8 and 9 are compared, it can be observed that there is a decrease in the bubble classes from 2.0 to 3.0 mm and an increase in smaller bubbles (0.5 to 2.0 mm). However, the changes are small and thus, SMD ($d_{S32} = 2.93/2.94$ mm) remains almost constant.

In order to study the influence of the superficial liquid velocity, another series of measurements with the same operating



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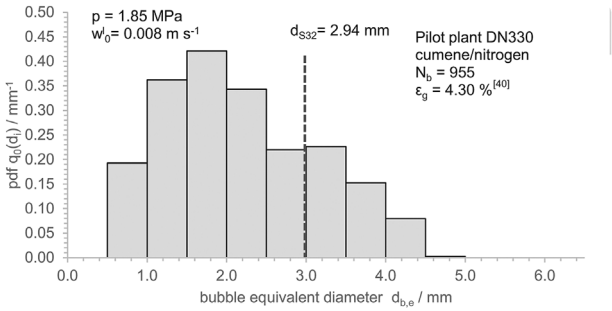


**Figure 9**. Bubble size distribution at p = 1.85 MPa, T = 71 °C with $w_0^g = 0.006\ \mathrm{m \cdot s^{-1}}$ and $w_0^l = 0.008\ \mathrm{m \cdot s^{-1}}$.

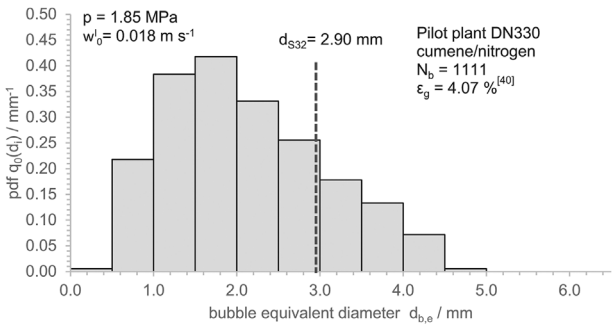


**Figure 10**. Bubble size distribution at p = 1.85 MPa, T = 71 °C with $w_0^g = 0.006\ \mathrm{m \cdot s^{-1}}$ and $w_0^l = 0.018\ \mathrm{m \cdot s^{-1}}$.

conditions is examined. In Figure 10 the parameters are the same as in Figure 9, except the superficial liquid velocity. A comparison of Figures 9 and 10 shows that the BSD is almost unaltered. However, the superficial liquid velocity is much smaller than the bubble rise velocity and the influence of the liquid velocity is assumed to be negligible.

In all BSDs shown in Figures 5–10, SMD remains almost constant, although differences are found in some bubble classes. However, the right-hand side of the BSD (larger bubble diameter) is not influenced significantly. Since the volume depends more strongly on the bubble diameter ($\sim d^3$) than the surface ($\sim d^2$), this nonlinear inter-relation leads to the fact that the SMD is more strongly influenced by larger bubbles, which has already been shown by Lapidus and Elgin.[39] Since the interfacial area for mass transfer is rather dominated by smaller bubbles because of the higher surface-to-volume ratio and residence time, it is clear that the SMD is an insufficient parameter to characterize bubbly flows and that optical analyses open a wide range for further investigations.

## Bubble Velocity Distribution

Besides the influence of industrial operating conditions on the BSD, their influence on the bubble rising velocity is almost unclear, no matter how important it is for the hydrodynamics and mass transfer performance of bubble column reactors. The absolute bubble rise velocity can be obtained with the EBIV as described above. Since the rising velocity is defined by the vertical component,[5] for the following descriptions only the vertical absolute bubble velocity is used. In order to quantify the bubble rise velocity only those bubble classes are considered for which 100 or more bubbles are available.

Figure 11 shows the absolute bubble velocity for the same bubble size classes that have been used for the BSD measurements. It can be observed that the bubble velocity for a pressure of 0.1 MPa is in general higher than for a pressure of 1.85 MPa.

Assuming that the influence of elevated pressures on BSD and SMD is too weak to cause a significant change in the rising velocity, the observed deviations can be referred to other influences, as will be discussed in the following section.

## Discussion

A deeper insight into the hydrodynamics of bubble column reactors under elevated pressures is possible with the previously

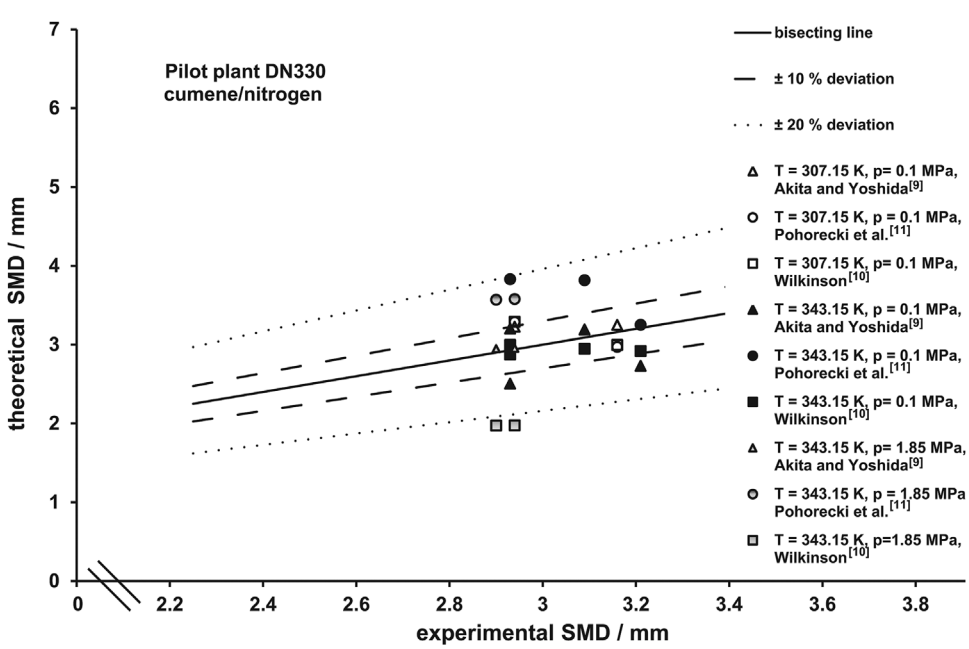


**Figure 11**. Bubble velocity distribution for two pressures measured in the material system cumene/nitrogen.



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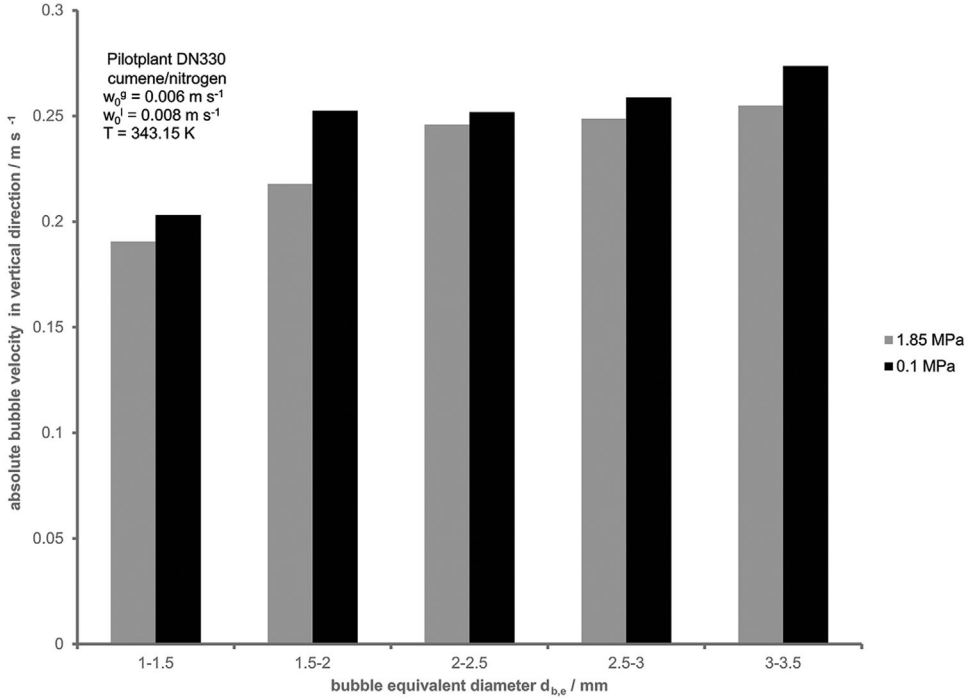


**Figure 12**. Comparison of theoretically and experimentally determined Sauter mean diameters.

presented results. It has been shown that with elevated pressure the BSD is shifted slightly to smaller bubble sizes, while in contrast the mean Sauter diameter is not influenced significantly (compare Figures 5–10).

Figure 12 shows how traditionally used correlations are able to predict the SMD for these industrial operating conditions. The SMDs theoretically determined by the correlations given in Equations (6–8) are plotted versus the experimentally determined SMDs.

The deviation between theoretical and experimental values depends on which correlation is considered for the particular operating conditions. Although the three different correlations differ in their predicted values, the majority of the values are within the 10 % range. The correlation according to Akita and Yoshida[9] shows the best agreement overall. This is not the case as regards the correlation by Wilkinson[10] because it underestimates the bubble diameter for elevated pressures.

As concerns the influence of elevated pressures on the rising velocity, a closer look at Equation (9) shows as influencing parameters the bubble size, the gas and liquid density, as well as the drag coefficient of the bubble. Assuming that the influence of elevated pressures on BSD and SMD is too weak to cause a significant change in the rising velocity, the observed deviations (compare Figure 11) should be referred to the density difference or the drag coefficient. It is well known that pressure leads to a compression of the gaseous phase that will increase the density of the bubble and reduce the buoyancy force as well as the rising velocity of the bubble. If one calculates the influence of the pressure on the density difference (see values in Table 2), it becomes obvious that the density difference for $T = 35\ ^\circ C$ is only lowered from 866 $kg \cdot m^{-3}$ to 844 $kg \cdot m^{-3}$, which causes a reduction in the rising velocity of less than 2 %. In addition, the drag coefficient depends strongly on the shape of the bubble which is dominated by the interfacial tension force and the external forces due to the movement. Comparing the interfacial tension for different pressures (see Table 2) it becomes clear that the interfacial tension at a temperature of $T = 70\ ^\circ C$ decreases from 26 $mN \cdot m^{-1}$ at ambient pressure to 21.7 $mN \cdot m^{-1}$ at $p = 1.85$ MPa. This decrease in interfacial tension can enable stronger shape deformations and leads to an increased drag coefficient. These are the obvious influencing parameters according to Equation (9), which cannot explain the entire reduction in a satisfactory way. Therefore, additional effects by increasing the pressure need further investigation. For example, increasing gas viscosity due to higher pressure could lead to a weaker internal circulation and decreasing bubble velocity.

**Table 2**. Physical properties of cumene/nitrogen

| Temperature (°C) | Pressure (MPa) | Density of liquid[1] ($kg \cdot m^{-3}$) | Density of gas[1] ($kg \cdot m^{-3}$) | Density difference ($kg \cdot m^{-3}$) | Interfacial tension[2] ($mN \cdot m^{-1}$) |
|---|---|---|---|---|---|
| 35 | 0.1 | 867.16 | 1.13 | 866.02 | 26 |
| 35 | 1.85 | 869.41 | 25.03 | 844.38 | 25.5 |
| 70 | 0.1 | 825.6 | 0.981 77 | 824.62 | 26 |
| 70 | 1.85 | 828.5 | 25.03 | 803.47 | 21.7 |

[1]Nist Chemistry WebBbook
[2]measured by Eurotechnica GmbH, Bargteheide



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While the discussion shows that the influence of elevated pressures on gas-liquid flows is most notably detectable by a decrease in bubble rise velocity, the BSD is almost unaffected. This reduced bubble rise velocity will directly cause an enhanced gas hold-up according to the two-fluid model:[39]

$$w_{bs,rel} = \frac{w_0^g}{\varepsilon_g} - \frac{w_0^l}{(1-\varepsilon_g)} \tag{13}$$

which is applicable to the homogeneous flow regime. Indeed, this dependency between pressure and gas hold-up has been reported in the literature[18] and measured for bubble column reactor by Rollbusch et al.[39] mentioned in this paper. From a quantitative point of view, the bubble rise velocity is reduced by 7.9 % (weighted average), whereas the gas hold-up is increased by 8.1 %,[40] which is in good agreement with the two-fluid model.

## CONCLUSIONS

The prediction of the gas hold-up in industrial bubble column reactors is still challenging and its dependency on pressure is controversially discussed. In this study investigations concerning the influence of pressure on BSD and bubble velocity in an industrial pilot plant have been conducted using cumene/nitrogen at elevated pressure of $p_{max} = 1.85$ MPa and temperatures of $T_{max} = 70$ °C for gas hold-ups up to 16 %. In order to be able to measure in bubble column reactors under such industrial conditions, the EBIV is introduced as a new measuring method.

It turns out that no significant influence of pressure on BSD can be identified, whereas the vertical absolute bubble velocity decreases with increasing pressure. A closer view shows that one reason for this circumstance is the decreasing density difference between the compressible gaseous phase and the incompressible liquid phase, resulting in a lower buoyancy force. In the end this effect does not entirely explain the reduction of the bubble velocity. Additional studies on further effects are required. Due to the reduction in bubble velocity, the increase of the gas hold-up at elevated pressures, that has been reported by Rollbusch et al.,[40] becomes explainable.

The results underline the importance of the bubble velocity with regard to the prediction of the gas hold-up in bubble column reactors. However, only a small number of correlations for predicting the gas holdup take the bubble velocity into account. This might also be due to the fact that the acquisition of the bubble velocity is hardly possible. Therefore this contribution shows a way to determine not only the bubble size but also the bubble velocity under industrial conditions. In turn, this information can be used for a more rigorous modelling of the gas hold-up, which considers the bubble velocity, especially under industrial conditions.

## ACKNOWLEDGEMENT

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## NOMENCLATURE

| | |
|---|---|
| $c_d$ | drag coefficient |
| $D$ | (inner) reactor diameter (m) |
| $d_b$ | bubble diameter (m) |
| $d_{b,e}$ | bubble equivalent diameter (m) |
| $d_o$ | diameter of the orifice |
| $d_{S32}$ | Sauter mean diameter (m) |
| $d^*$ | dimensionless bubble diameter according to Equation (4) |
| $g$ | gravity constant ($m \cdot s^{-2}$) |
| $H$ | height (m) |
| $n$ | constant |
| $N_b$ | number of evaluated bubbles |
| $n_i$ | number of bubbles in class "i" |
| $p$ | pressure (Pa) |
| $\Delta p$ | difference of pressure in the dispersed state (Pa) |
| $\Delta p_0$ | pressure difference of the liquid height (Pa) |
| $q_0$ | probability density function of number ($m^{-1}$) |
| $S$ | surface of an ellipsoid ($m^2$) |
| $T$ | temperature (°C) |
| $V$ | volume ($m^3$) |
| $w_0^g$ | superficial gas velocity ($m \cdot s^{-1}$) |
| $w_0^l$ | superficial liquid velocity ($m \cdot s^{-1}$) |
| $w_b$ | single bubble velocity in stagnant liquid ($m \cdot s^{-1}$) |
| $w_{bs,abs}$ | absolute bubble velocity in a swarm ($m \cdot s^{-1}$) |
| $w_{bs,rel}$ | relative bubble velocity in a swarm ($m \cdot s^{-1}$) |
| $w_l$ | local liquid velocity ($m \cdot s^{-1}$) |
| $w_o$ | velocity in the orifice ($m \cdot s^{-1}$) |
| $\varepsilon_g$ | gas hold-up |
| $\sigma$ | interfacial tension ($mN \cdot m^{-1}$) |
| $\rho_g/\rho_l$ | gas/liquid density ($kg \cdot m^{-3}$) |
| $\Delta\rho$ | density difference of gas and liquid phase ($kg \cdot m^{-3}$) |
| $\eta_l$ | dynamic viscosity of the liquid phase ($Pa \cdot s$) |
| $\nu_l$ | kinematic viscosity of the liquid phase ($m^2 \cdot s^{-1}$) |
| $\Phi$ | critical parameter according to Equation (3) |
| Fr | Froude number $= Fr = \frac{w_o^2}{d_o g}$ |
| We | Weber number $We = \frac{w_o^2 d_o \rho_g}{\sigma}$ |

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